SURVEYING & LEVELLING · NOTE SL14

Job Layout — Grade Stakes, Contour Bunds, Graded Terraces, Foundations and Earthwork

Course topics: IDE5.5 L9, SWCE4.5 L10, SWCE4.5 L11

Purpose

Job layout transfers the engineering plan to the ground: it locates the works, lines, grades and elevations needed for construction. Every engineering work needs it before building starts.

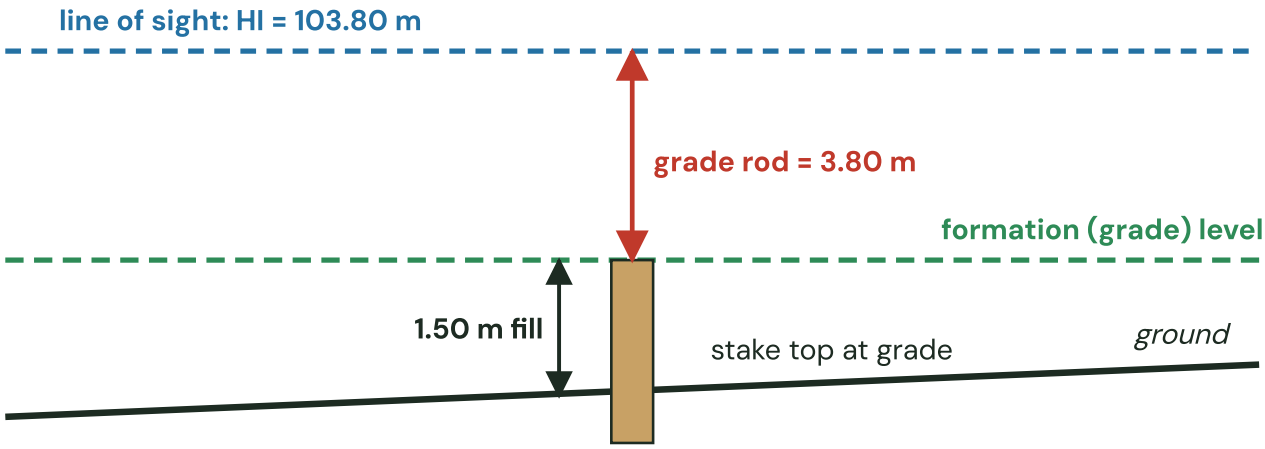
Setting grade stakes

In land grading, grade stakes show the cut or fill needed to bring the ground to the planned grade. After drawing the grade line on the profile, compute the grade (formation) level of each station and the grade rod reading:

$$\text{Grade rod reading} = \text{HI} - \text{formation level}$$

Method 1 — top of stake at grade

Formation level at P = 100.00 m; HI = 103.80 m; ground level at P = 98.50 m.
Grade rod = 103.80 - 100.00 = 3.80 m.
1. Drive the stake partly in at P and read the rod on its top.
2. Drive it deeper until the reading is exactly 3.80 m.
3. The stake top now stands 100.00 - 98.50 = 1.50 m above the ground — the fill needed at P.



Driving a grade stake until its top is at formation level

Method 2 — blue-top levelling

- Mark two lines on each stake: **blue** for formation level and **red** for ground level.
- Fill points: blue above red. Cut points: blue below red. The gap between them is the cut or fill.

- The grade is reached when the worked surface is flush with the blue line. This method is common in farm work.

Method 3 — stake with a cut or fill mark

- Drive the stake only to locate the point and write the depth of cut or height of fill on it — e.g. **C 8** = cut of 8 units, **F 12** = fill of 12 units.
- Use **short stakes for cuts and long stakes for fills** (mark the fill depth on long stakes), so operators see the cut and fill areas at a glance.
- Build fills 10–15% higher than marked to allow for settling.

Laying out a contour bund or level terrace

1. Decide the vertical interval for the first bund and set up the level.
2. Take a rod reading on the hilltop and fix the height of the first bund.
3. Send the rodman up or downhill until the rod shows the required reading; drive a stake there.
4. Move the rodman about 15 m along the contour and guide him until the same reading is obtained; stake it.
5. Repeat until the bund line is complete, shifting the instrument forward when needed but keeping the same rod reading.
6. Set the second bund the same vertical interval below the first.

Adjust a few stakes slightly to smooth sharp turns, keeping departures from the true grade to a few centimetres.

Laying out a graded terrace

1. Decide the vertical interval for the top terrace and start staking from the **outlet**.
2. Place the first stake at the outlet.
3. Place the next stake 15 m up the channel, higher than the previous one by the channel grade × 15 m.
4. Continue every 15 m; at sharp turns reduce the interval to 7.5 m.
5. Shift stakes slightly to smooth the alignment; keep the channel grade by cutting or filling.

> Channel grade 0.4%: rise per 15 m = 0.004 × 15 = 0.06 m. If the rod reads 5.60 m at the second stake, it should read 5.60 - 0.06 = 5.54 m at the third stake 15 m away.

> **Check this figure.** The source notes place the second stake "15 m up the channel, 20–25 cm higher". At the grades used for terrace and graded-bund channels — 0.2–0.3% in the ICAR eCourse, 0.4% in the example above — a 15 m step rises only about 3–6 cm. "20–25 cm" is most likely 20–25 mm; always compute the rise as grade × distance.

Laying out a building foundation

1. Start from side AB and corner C; locate corner D 24 m from C along AB.
2. Erect perpendiculars at C and D and measure 12 m on each to locate E and F.
3. Check the rectangle by measuring the diagonals CE and DF and the side FE.
4. From E measure 6 m along FE to locate I, and a further 12 m to locate J.
5. Erect perpendiculars at I and J and measure 6 m to locate G and H.
6. Check the diagonals GI and HJ and the side GH.

Earth quantities — end-area method

$$V = l \times (A_1 + A_2) / 2$$

V = volume of excavation or fill (m^3); l = distance between successive cross-sections (m); A_1, A_2 = their end areas (m^2).

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra (IIT Kharagpur), reviewed by Dr B. C. Mal; Lesson 6.2 (contour and graded bunds). ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Grade rod reading equals

(a) HI + formation level
(b) HI - formation level
(c) formation level - ground level
(d) ground level - HI

Q2. With HI = 103.80 m and formation level = 100.00 m, the grade rod is

(a) 1.50 m
(b) 3.80 m
(c) 5.30 m
(d) 103.80 m

Q3. In blue-top levelling, at a fill point the blue line is

(a) below the red line
(b) above the red line
(c) at ground level
(d) absent

Q4. Fills are built higher than marked by about

(a) 1–2%
(b) 10–15%
(c) 25–30%
(d) 50%

Q5. Stakes for a graded terrace are set every

(a) 5 m
(b) 15 m
(c) 30 m
(d) 50 m

Q6. For a 0.4% channel, rod readings on successive stakes 15 m apart differ by

(a) 0.006 m
(b) 0.06 m
(c) 0.6 m
(d) 6 m

Answers

Q1 — (b) HI - formation level
Q2 — (b) 3.80 m
Q3 — (b) above the red line
Q4 — (b) 10–15% — *To allow for settling.*
Q5 — (b) 15 m — *7.5 m at sharp turns.*
Q6 — (b) 0.06 m